

I. Project Title

StudyUp: A Smart Task Prioritization Tool for Students

II. Problem Statement

Many students are struggling to manage their school work due to overwork, therefore, developing burnout. Also, due to having to manage work from multiple subjects, they find it hard to prioritize assignments. This problem has many effects, one, is not meeting the deadline for work, leading to a lot of stress and often to poor academic performance. The goal of StudyUp is to help students manage their work load by prioritizing their tasks on the basis of due dates and estimated work time required for each task.

III. Project Objectives

1. Prioritize: Automatically sort a list of student tasks by urgency (due date).
2. Track: Provide a clear view of remaining tasks to help the student visualize their daily workload.
3. Calculate: The sum of estimated hours for all unfinished tasks in order to inform study time required for completion of all schoolwork assigned.

IV. Planned Features

- Task Entry: Users can input task names, due dates, and estimated hours to complete.
- Sorting Algorithm: The system will sort tasks from "most urgent" to "least urgent."
- Summary View: Display a formatted list of tasks showing the priority order and total time remaining.

V. Planned Inputs and Outputs

Inputs:

- Task Name (String)
- Due date (Date/Number of days remaining)
- Estimated Time to Complete (Hours)

Outputs:

- Sorted list of tasks.
- Total time required to finish all tasks.
- A "Top Priority" alert for the task due soonest.

PsuedoCode

START

Ask the student how many tasks they have, and get the name, due date, and hours for each one.

Go through the list of tasks.

As long as there are still tasks left to check:

Compare two tasks - if one is due sooner but listed after the other, swap them.

Move on to the next task.

Keep doing this until everything is in order from the earliest due date to latest.

Show the sorted list of tasks. Tell the student which task is most urgent. Add up the hours for everything left to do, and show the total.

END